CSA0712-COMPUTER NETWORK

Security and Layer 2 Optimization

Defensive Configuration of Switchports Connecting the Server

Presented by:krishnapriya P

(192511279)

Introduction

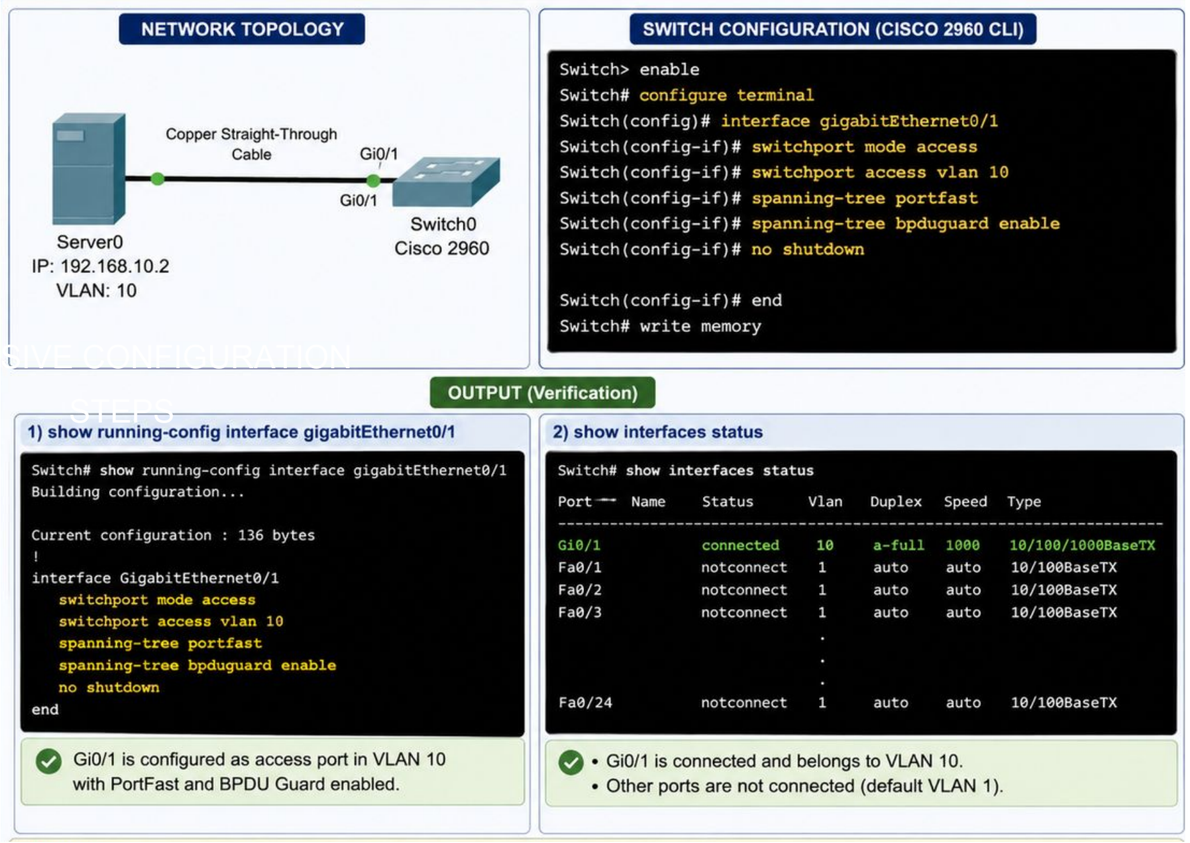


What is Layer 2 Security?

- Layer 2 security protects the switching infrastructure.
- Prevents unauthorized access and network attacks.
- Ensures reliable communication between servers and switches.
- Uses secure switchport configurations to improve performance and availability.

Defensive Configuration Steps

- Configure the port as an Access Port
- Assign the correct VLAN
- Enable PortFast
- Enable BPDU Guard
- Disable unused interfaces
- Place unused ports in a Dead VLAN
- Use strong administrative passwords



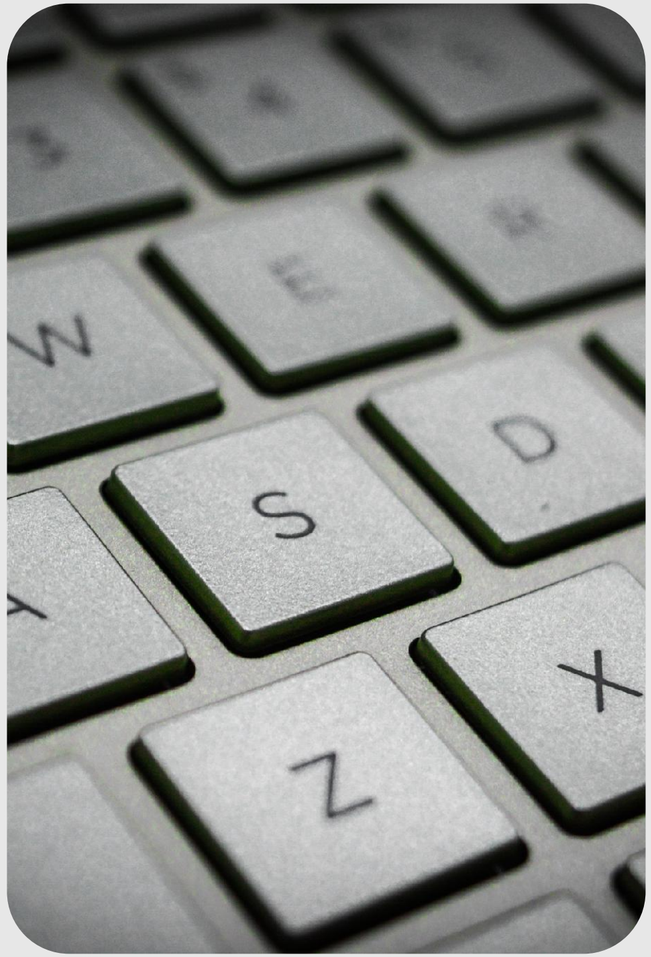
Why Use Spanning-Tree PortFast?

PURPOSE OF PORTFAST:

- Allows the port to move directly to the Forwarding state.
- Eliminates the normal STP delay (30–50 seconds).
- Servers become available immediately after startup.
- Improves application and network performance.
- Recommended only for end devices such as:
- Servers
- PCs
- Printers

Benefits

- Faster connectivity
- Reduced boot time
- Improved user experience





PortFast with BPDU Guard

PortFast should always be combined with BPDU Guard.

If a switch is accidentally connected:

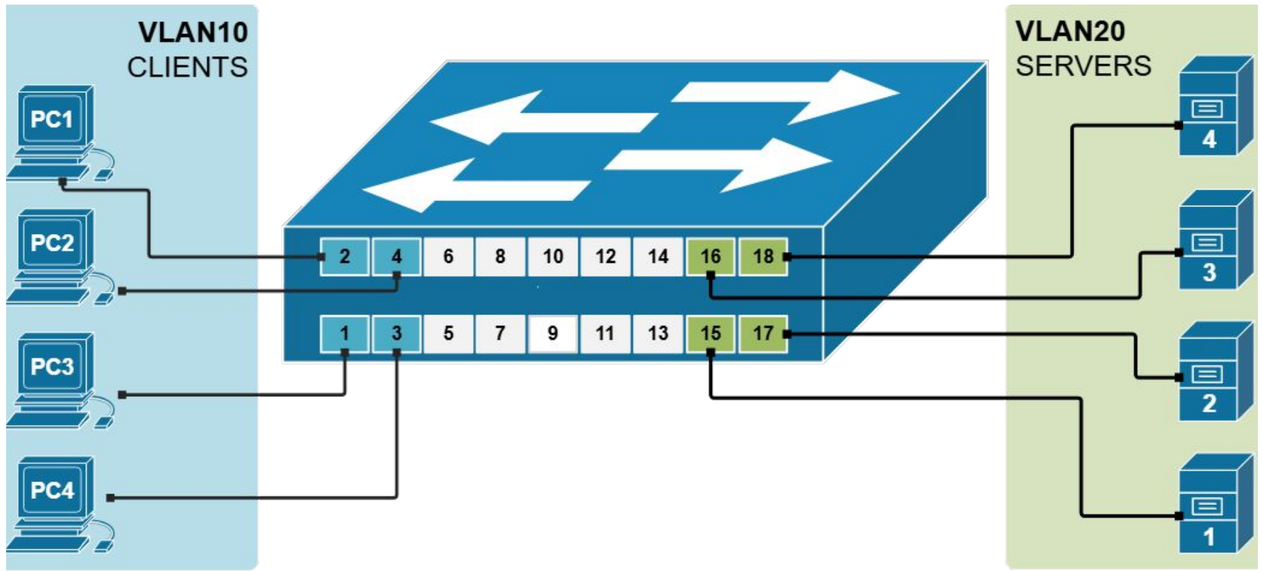
- BPDU Guard detects BPDU packets.
- Automatically disables the port.
- Prevents switching loops.
- Protects the network from accidental or malicious connections



Risks of Unused Switch Ports

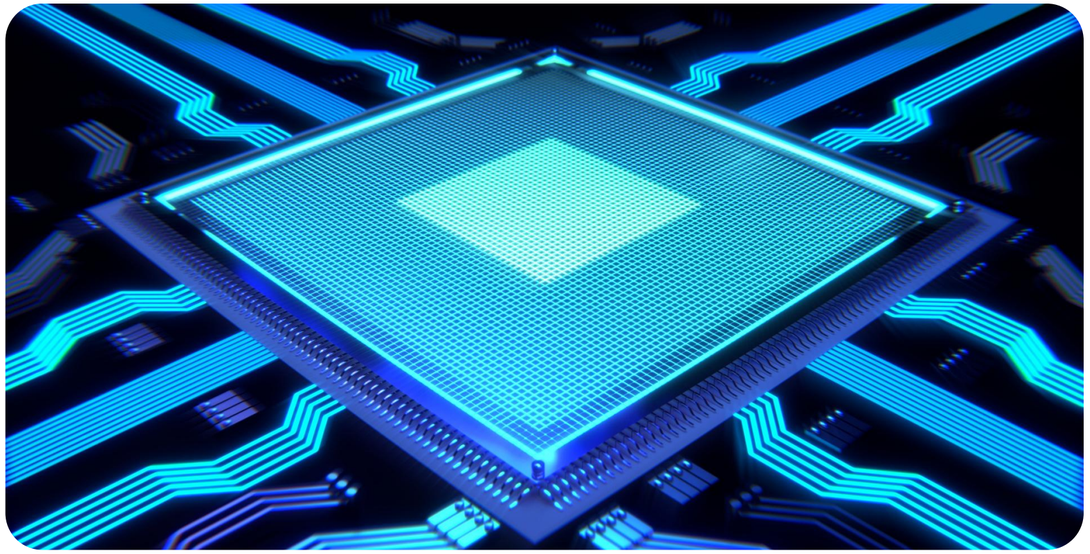
- Unauthorized network access
- MAC flooding attacks
- Rogue devices
- VLAN hopping attacks
- Network loops
- Data theft
- Malware entry

Dead VLAN



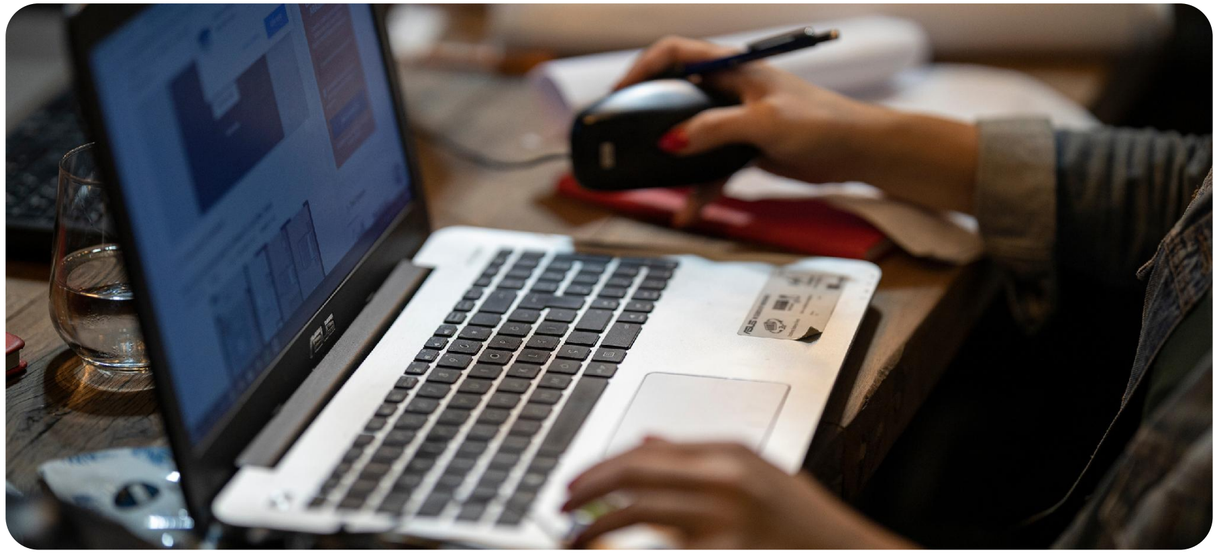
Advantages:

- Prevents unauthorized access
- Isolates unused ports
- Improves Layer 2 security
- Simplifies network management



Conclusion

- PortFast reduces connection delay for servers.
- BPDU Guard protects against switching loops.
- Unused ports should never remain active.
- Assigning unused ports to a Dead VLAN increases security.
- Proper Layer 2 optimization improves both




THANK YOU